

Technology-Driven Sports Digital Twin: Human-Machine Collaborative Olympic Strategic Decision System

Abstract—In the era of digital transformation, sports increasingly rely on intelligent technologies. This study proposes a strategic decision-making framework for the Olympic Games that integrates statistical models, machine learning, and digital twin architecture. Organized into data, model, and application layers, the system fuses historical and real-time data to support adaptive planning. To address cross-national disparities, the framework employs k-means clustering, time series forecasting, and Markov chains, incorporating host nation and elite coach effects. A microservice-based digital twin enhances predictive accuracy and system robustness. To mitigate data sparsity and ensure transparency, data augmentation and interpretable models are utilized. Empirical findings indicate the host nation effect yields a 9.52% increase in medal counts, while middle-income countries benefit from targeted training and cost-efficient strategies. The framework also forecasts the 2028 Olympic medal distribution. It presents a novel, ethically grounded paradigm for human-machine collaboration in sports analytics.

Index Terms—K-means clustering, Analytic Hierarchy Process, Markov model, machine learning, digital twins

I. INTRODUCTION

The rapid development of data science, AI, and digital twin technologies reshapes sports strategy through intelligent, data-driven decision-making. As the pinnacle of global competition, the Olympic Games reflect not only athletic strength but also national strategy and economic capacity. Accurate medal prediction, grounded in data ethics and algorithmic transparency, is increasingly vital. While traditional statistical models remain valuable, machine learning and multi-source data offer enhanced accuracy and actionable insights.

Although previous studies advance Olympic medal prediction, key limitations persist. Bernard et al. [1] use Logit models based on population and GDP, while Schlembach et al. [2] apply random forests to validate these factors. Zhu et al. [3] highlight the value of data-driven prediction and strategy. However, most studies focus on socioeconomic indicators, overlooking cross-national differences. Unified models struggle to reflect diverse country profiles, and issues like data sparsity and algorithmic transparency remain underexplored—especially for smaller nations. The potential of human-machine collaboration also remains largely untapped.

This study addresses these gaps by developing an Olympic strategic decision-making system that integrates statistical models, machine learning, and digital twins. By combining historical and real-time data, the framework enables more precise, adaptive, and fair predictions of national performance.

II. SYSTEM CONSTRUCTION

Integrating traditional statistical models with modern machine learning techniques enhances Olympic medal prediction and training strategy optimization.

The research divides into two phases:

- 1) Foundational prediction and strategy exploration
- 2) Multi-model integration and digital twin platform

The following sections outline the methodologies and system architecture across each phase, focusing on the data and model layers, with selective attention to the application layer.

A. Hierarchical Strategy for Phase 1

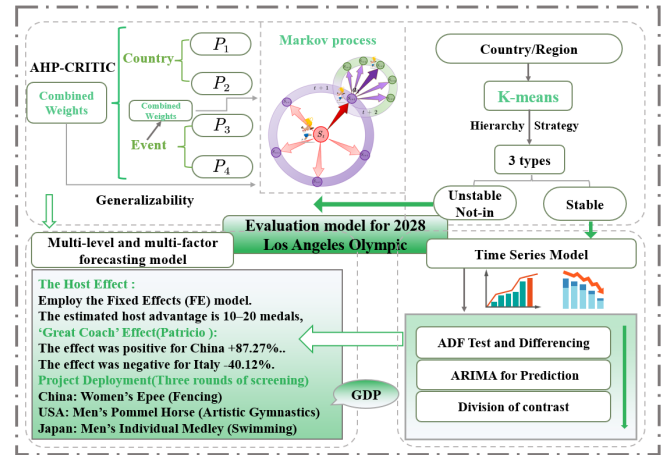


Fig. 1. The structure of phase 1

To address disparities in national sports development, we adopt a hierarchical prediction strategy based on country classification.

At the data layer, we gather information from the official Olympic website, including athlete profiles, host records, medal counts, and event results. The data are cleaned and standardized, with a particular focus on ensuring consistency in the NOC field to enable accurate national comparisons.

At the model layer, we assess performance consistency across nations using the Coefficient of Variation (CV), defined as

$$CV = \frac{\sigma}{\mu}, \quad (1)$$

where σ and μ denote the standard deviation and mean. Figure 2 shows the CV heat map.

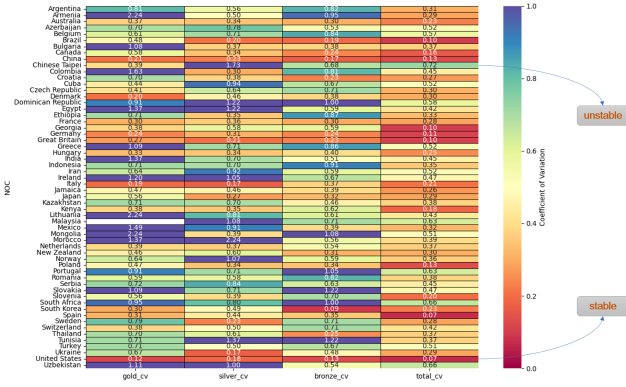


Fig. 2. Coefficient of Variation (CV)

We use K-means clustering to group countries by medal performance. Each point is assigned to the nearest centroid:

$$C_i = \arg \min_j (\|x_i - c_j\|^2), \quad (2)$$

and centroids are updated as:

$$c_j^{\text{new}} = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i. \quad (3)$$

Three clusters prove optimal. Cluster 2 countries, with consistent participation, are labeled “stable”; others are “unstable.” Figure 3 shows the visualization.

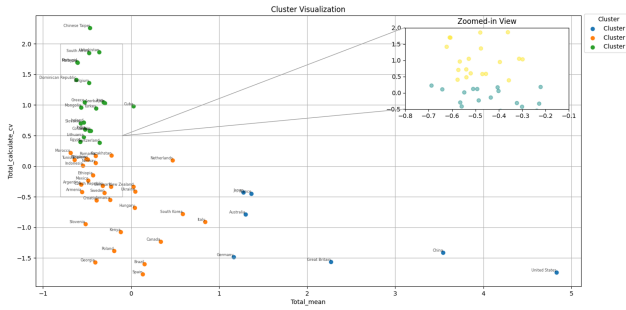


Fig. 3. K-means Clustering Visualization

B. Time Series and AHP-Markov Model

For countries with stable historical performance, medal counts show strong cyclical and trend patterns, which time series models capture effectively. We begin by using the ADF test to check for stationarity. If the data are non-stationary, first-order differencing is applied. We then fit and predict using the ARIMA (1,1,1) model, which captures past medal trends, and the predicted values are restored to the original scale.

For countries with high medal volatility or limited historical data, traditional time series models prove less effective. We adopt a Markov strategy as shown in Figure 4, focusing only on the last two Olympic Games to capture instability.

To comprehensively analyze medal prediction, we consider the four transition probability matrices that combine transition probabilities at both the country and event levels using the

Analytic Hierarchy Process (AHP). AHP is used to derive the weight vector W_1 :

$$W_1 \approx [0.054, 0.166, 0.191, 0.589], \quad (4)$$

for prediction fusion.

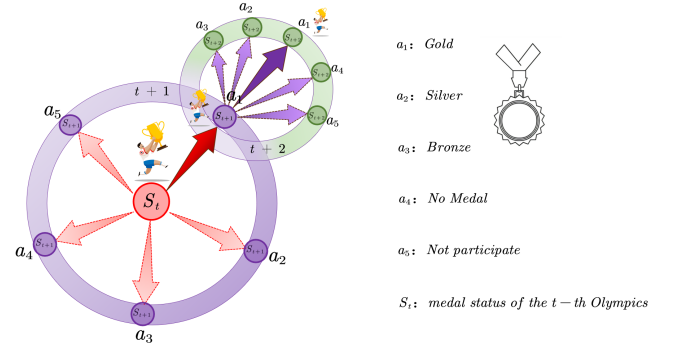


Fig. 4. Schematic diagram of Markov process

C. AHP-Fusion and Microservice Digital Twin of Phase 2

In Phase 2, we start with a combination of real-time data, using the Internet of Things (IoT), sensors, and social media to gather athlete training data, game dynamics, spectator sentiment, and more. Then we integrate the models from Phase 1 into a unified ensemble framework and develop a microservice-based digital twin platform. Each model produces predictions $\hat{y}_i$ with uncertainty estimates. The final output is a weighted linear combination of these predictions:

$$\hat{y} = \sum_{i=1}^n w_i \hat{y}_i, \quad \text{with} \quad \sum_{i=1}^n w_i = 1, \quad (5)$$

where weights w_i are determined by cross-validation and AHP. To further improve performance, ensemble methods like stacking and residual-based boosting are used to minimize biases and enhance generalization.

The digital twin platform consists of three layers: the data layer aggregates historical and real-time data through ETL pipelines and APIs; the model layer hosts the integrated models as microservices for scalable deployment; and the simulation and visualization layer provides an interactive dashboard for scenario analysis and model output visualization.

Deployed on a cloud native infrastructure with Docker and Kubernetes, the platform ensures flexibility and scalability. It not only replicates real-world Olympic dynamics but also allows for virtual policy testing and strategy optimization. Users monitor model performance, adjust parameters, and run simulations, making the platform a powerful tool for data-driven Olympic forecasting.

D. Extended Influencing Factors

Host countries often exceed their Olympic averages thanks to better infrastructure, stronger athlete support, and targeted investment. This “host effect” helps guide medal prediction and resource planning [4]. We use a Fixed Effects (FE) model

and Least Squares Dummy Variable (LSDV) method to control for country traits and time trends, estimating how hosting affects medal counts.

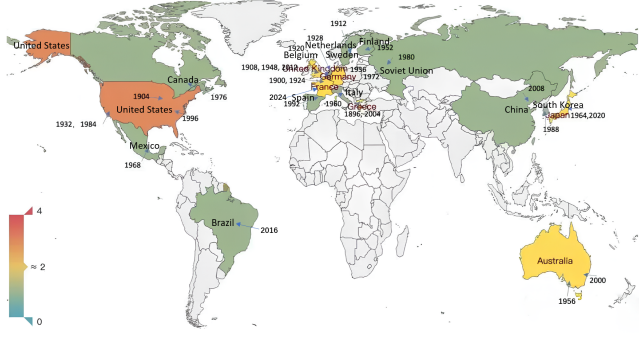


Fig. 5. Distribution of previous Olympic host countries

Unlike athletes, coaches move freely between countries, creating an “elite coach effect” [5]. For example, Patricio’s 2020 move from Italy to coach China’s women’s racewalk team has a clear impact on the results of both nations. We assess such effects and offer strategy suggestions for three pilot countries, with corresponding results presented in Section IV.

We also extend the SHAP method of Shi et al. [6] by adding GDP and population to improve the explainability of the model.

III. ETHICS AND FAIRNESS GUARANTEE

Ensuring ethics and fairness in the construction of the Olympic strategic decision system is crucial, especially in addressing data sparsity for smaller countries and enhancing model interpretability. To tackle data sparsity, the Synthetic Minority Over-sampling Technique is employed. It generates synthetic data by interpolating existing minority class samples, balancing the dataset, and improving the model’s predictive capability for smaller nations [7].

In Phase 1 of the AHP-Markov model, for countries with limited participation or scarce data, we estimate their potential by focusing on event-level performance through P_3 and P_4 matrices, reducing reliance on overall Olympic performance data.

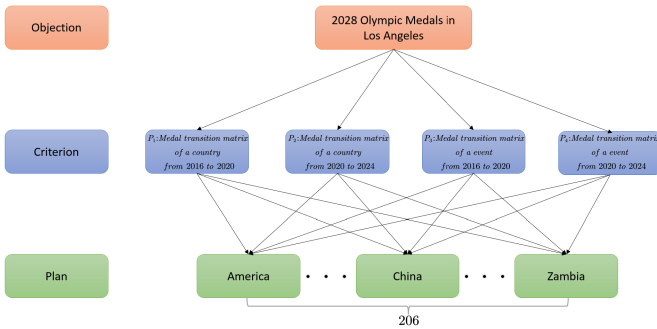


Fig. 6. Hierarchy of medal prediction model

To reduce model bias, fairness regularization terms are added during training to penalize uneven predictions across different groups. This ensures the model performs consistently for all groups. Fairness metrics, such as “Equal Opportunity” and “Equalized Odds,” are used to assess and improve fairness. “Equal Opportunity” ensures equal true positive rates, while “Equalized Odds” adjusts both true and false positive rates across groups [8].

For transparency and interpretability, LIME and SHAP are incorporated. LIME fits local linear models to explain individual predictions, helping users understand decision processes in complex models [9]. SHAP, based on Shapley values, evaluates each feature’s contribution to the final prediction, clarifying the model’s decision-making process [10].

These methods ensure that the Olympic strategic decision system is fair, transparent, and aligned with ethical standards.

IV. SYSTEM APPLICATION RESULTS DISPLAY

A. System Robustness Evaluation

We first evaluate the system’s performance using standard metrics. The predictive and explanatory power of the model is quantified by the coefficient of determination, R^2 , defined as

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \hat{y}_i)^2} \approx 0.879. \quad (6)$$

This result indicates that the model exhibits strong predictive performance.

B. Medal Prediction and Fairness

The system forecasts the medal distribution for the 2028 Los Angeles Olympics. It is important to note that the United States hosts the next Olympic Games. Our system analysis shows that the host effect contributes approximately 9.52% to the predictions.

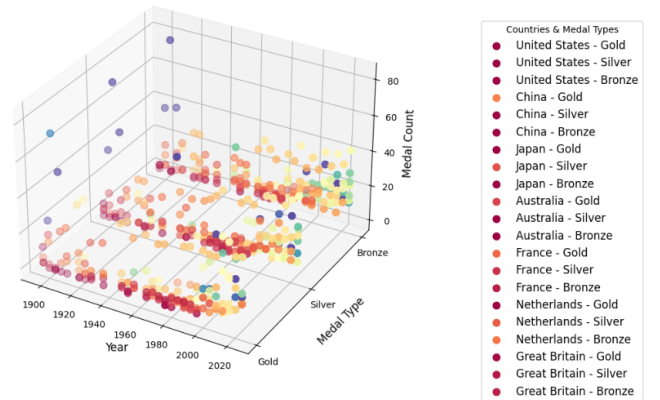


Fig. 7. Medal Counts for Countries

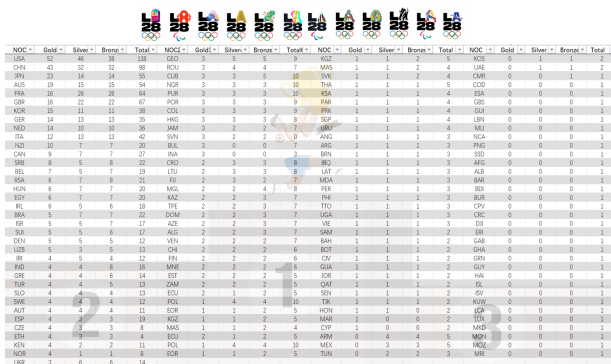


Fig. 8. Medal List 1

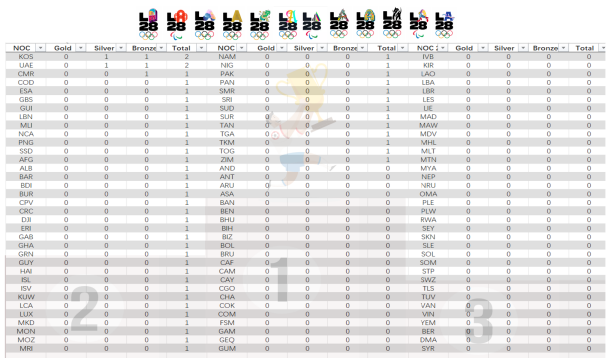


Fig. 9. Medal List 2

To ensure fairness, the model incorporates predictions for countries that historically do not secure any medals.

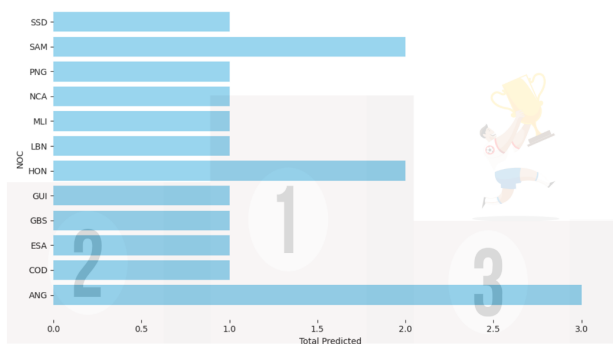


Fig. 10. First-time winner

C. Strategic Deployment

For Patricio's example, we calculate the following results. The Weighted Total Score (WTS) is calculated as:

$$\text{WTS} = 5 \times \text{Gold} + 2 \times \text{Silver} + 1 \times \text{Bronze}. \quad (7)$$

The transfer of Patricio has a statistically significant impact on both teams' performances:

- 1) China has a performance boost of +87.27%.

TABLE I
WEIGHTED TOTAL SCORES (WTS) FOR CHINA AND ITALY (2012–2024).

Country	Period	Gold	Silver	Bronze	WTS	Avg WTS
China	2012–2020	1	1	1	8	2.67
China	2024	1	0	0	5	5.00
Italy	2012–2020	1	0	0	5	1.67
Italy	2024	0	0	1	1	1.00

TABLE II
SIGNIFICANCE TESTING OF WEIGHTED TOTAL SCORES (WTS).

Team	Avg WTS Before	WTS After	<i>p</i> -value	Conclusion
China	2.67	5.00	$p < 0.05$	Increase
Italy	1.67	1.00	$p < 0.05$	Decline

- 2) Italy has a performance decline of -40.12%.

For the three pilot countries, our system provides targeted strategic recommendations:

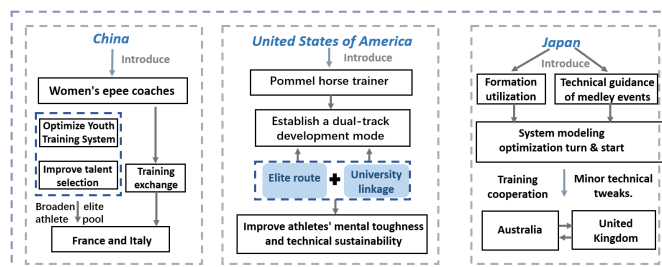


Fig. 11. Strategic deployment proposals for 3 pilot cities

D. Other Original Insights

Our analysis shows that GDP, population, and Olympic medal counts follow diminishing returns. Higher GDP and larger populations link to more medals, but the impact decreases beyond a certain point. While population size helps, as seen in China and the USA, it does not guarantee success in countries with poor infrastructure, such as India.

The host nation advantage partly results from economic and demographic factors. Although hosts typically win more medals, this is largely due to existing infrastructure and population size, not just hosting the games.

Lastly, the legacy effect suggests that continued investment in infrastructure and athlete development extends the host nation's advantage. Lower-income nations focus on specific sports, while wealthier countries invest more broadly across events.

V. CONCLUSION

This study develops an Olympic strategic decision system that combines statistical models, machine learning, and digital twin technology. By using historical and real-time data, the system provides a detailed and multi-level analysis of medal predictions. Methods like K-means clustering, time series analysis, and Markov processes help capture trends, cycles,

and uncertainties in sports development. The use of AHP and ensemble learning further improves prediction accuracy and offers country-specific strategy suggestions.

The study also focuses on fairness and transparency by applying techniques such as SMOTE, LIME, and SHAP to reduce data bias, especially for smaller countries. Overall, the system offers strong support for decision-making in sports and shows promise for future use in real-time updates and adaptive modeling.

VI. FUTURE WORK

Future work includes more real-time data sources like IoT sensors and social media to make predictions more accurate and responsive to changes. We also explore models that adjust and learn in real time to stay up-to-date with changes in the sports world. These areas of future research contribute to refining and expanding the current system, ultimately leading to more accurate, equitable, and actionable predictions for Olympic athletes and teams.

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